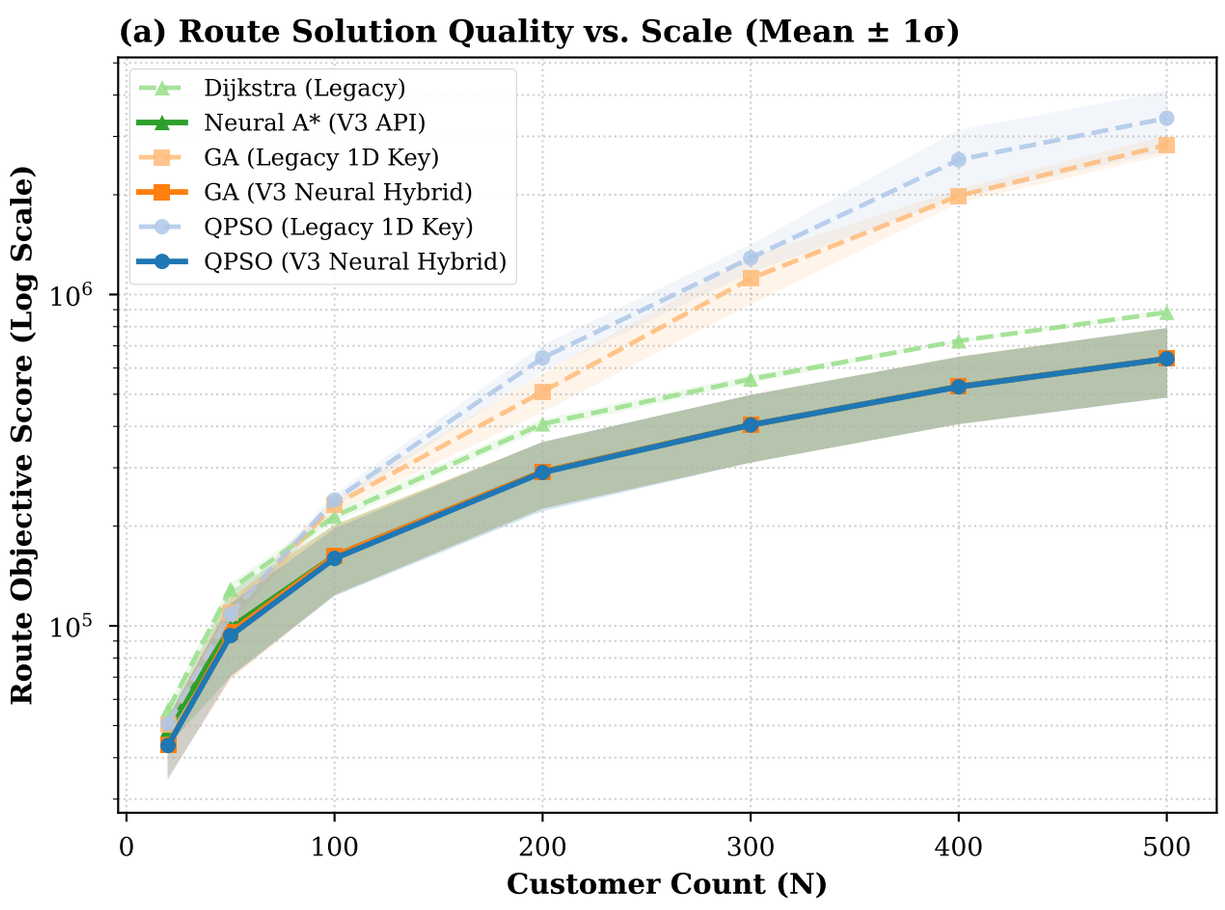
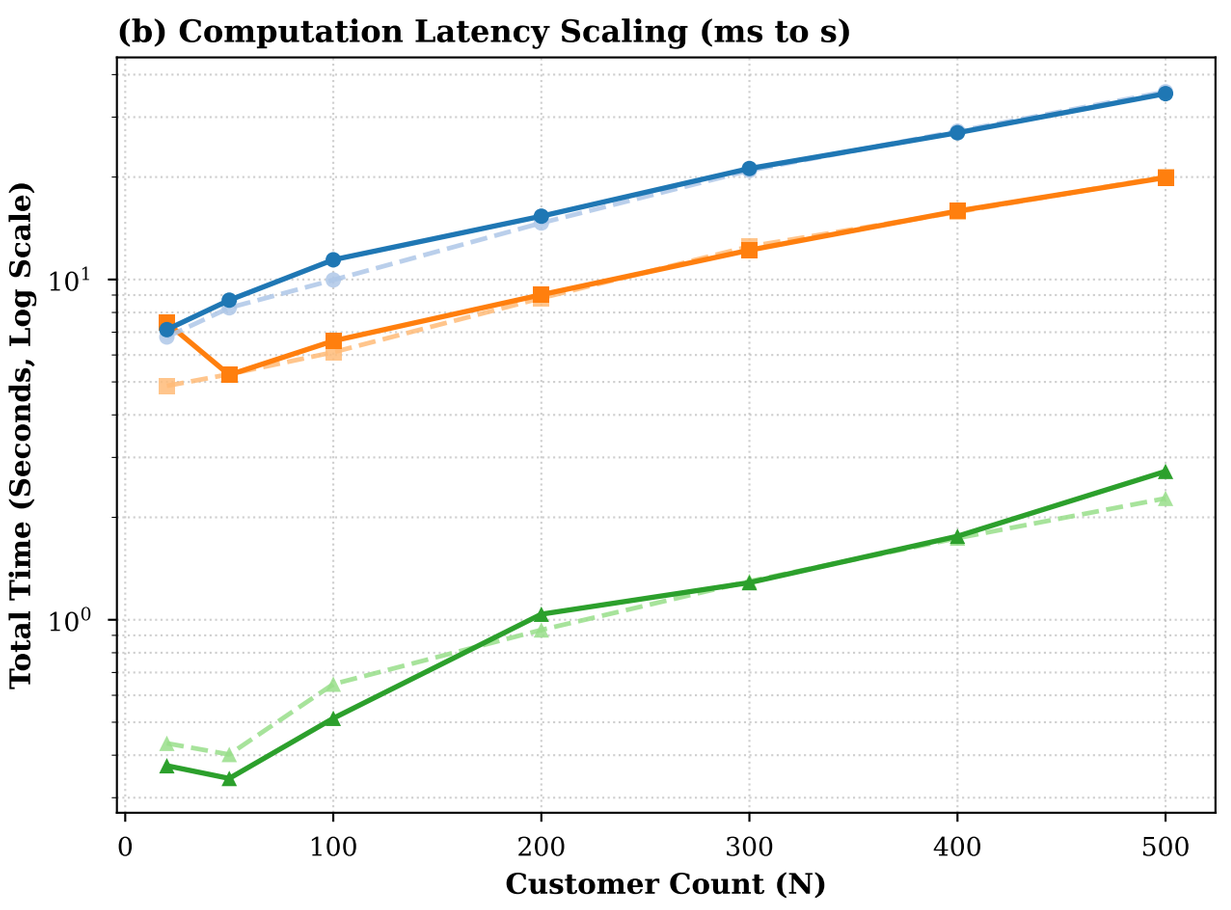
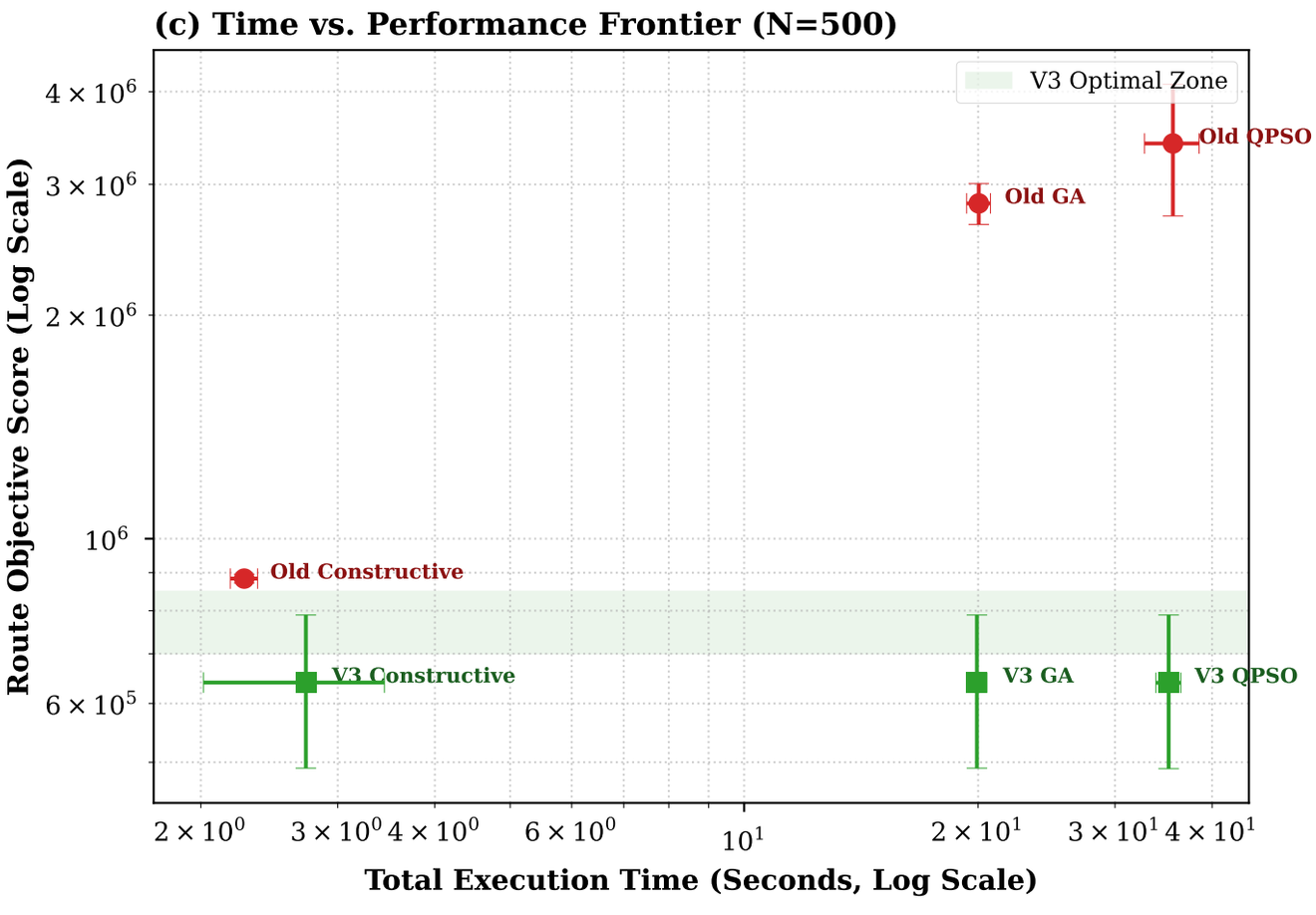
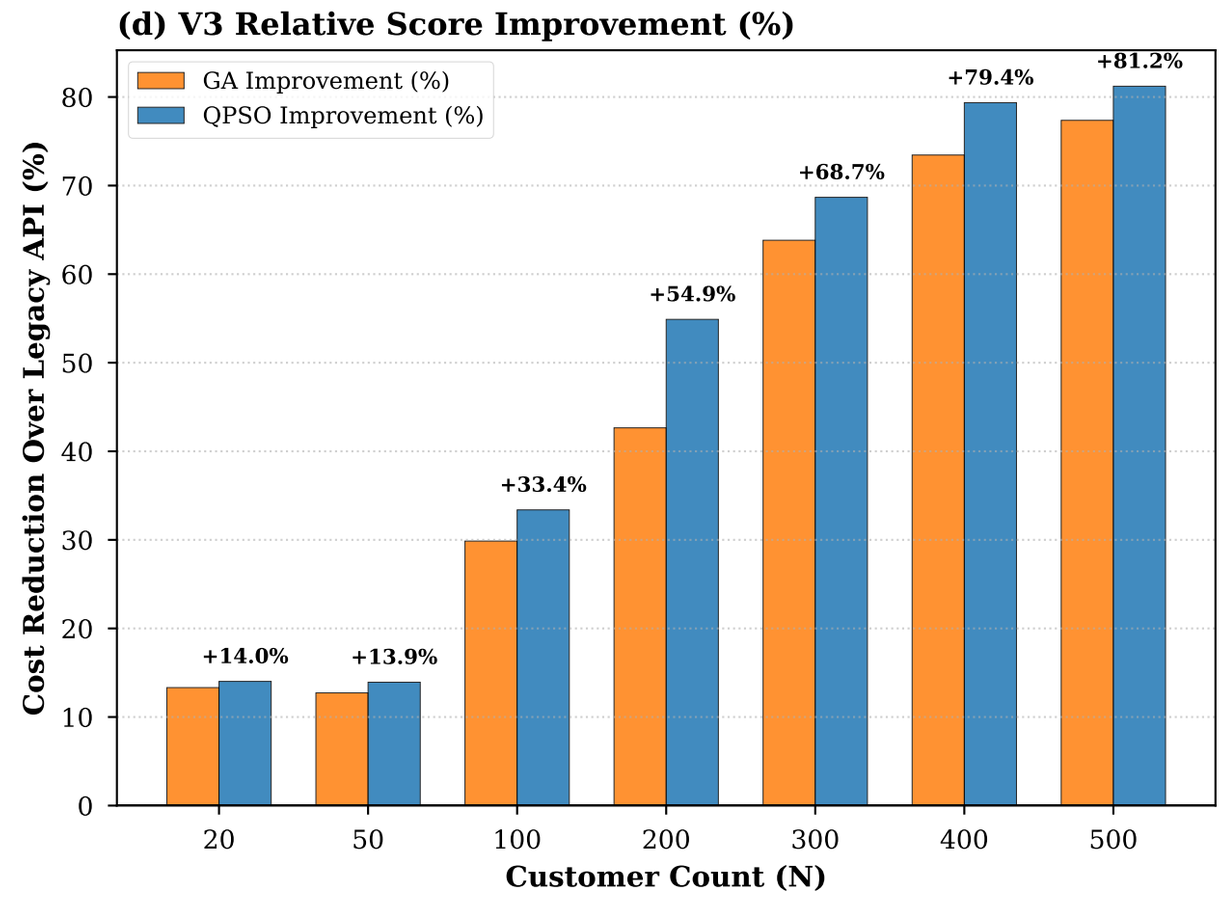

(a) Route Solution Quality vs. Scale (Mean ± 1σ)
Dijkstra (Legacy)
Neural A* (V3 API)
GA (Legacy 1D Key)
GA (V3 Neural Hybrid)
QPSO (Legacy 1D Key)
QPSO (V3 Neural Hybrid)
Route Objective Score (Log Scale)
10^6
10^5
0
100
200
300
400
500
Customer Count (N)


(b) Computation Latency Scaling (ms to s)
Total Time (Seconds, Log Scale)
10^1
10^0
0
100
200
300
400
500
Customer Count (N)


(c) Time vs. Performance Frontier (N=500)
V3 Optimal Zone
Old QPSO
Old GA
Old Constructive
V3 Constructive
V3 GA
V3 QPSO
Route Objective Score (Log Scale)
4×10^6
3×10^6
2×10^6
10^6
6×10^5
2×10^0
3×10^0
4×10^0
6×10^0
10^1
2×10^1
3×10^1
4×10^1
Total Execution Time (Seconds, Log Scale)


(d) V3 Relative Score Improvement (%)
GA Improvement (%)
QPSO Improvement (%)
Cost Reduction Over Legacy API (%)
+14.0%
+13.9%
+33.4%
+54.9%
+68.7%
+79.4%
+81.2%
0
10
20
30
40
50
60
70
80
20
50
100
200
300
400
500
Customer Count (N)